1. Create the virtual environment (use 3.12 explicitly)

```
py -3.12-64 -m venv venv
```

2. Activate it

```
venv\Scripts\activate
```

You should see (venv) appear at the start of your prompt.

3. Confirm it's using 3.12

```
python --version
```

Should print Python 3.12.0.

4. Install all dependencies

```
pip install -r requirements.txt
```

Wait for it to finish — takes 1–2 minutes.

5. Test each module one by one (run each, check no red error text appears)

```
python semantic_engine/ast_diff.py
python dependency_graph/dep_analyzer.py
python data/generate_dataset.py
python conflict_prediction/predict.py
python compliance_engine/compliance_check.py
```

6. Test the self-healing module (paste this whole block as one command)

```
python -c "from semantic_engine.ast_diff import diff_versions; from
self_healing.suggest import suggest_resolution;
a=open('sample_repo/branch_a.py').read();
b=open('sample_repo/branch_b.py').read(); [print(suggest_resolution(c)) for c in
diff_versions(a,b)]"
```

7. Launch the full dashboard (your main demo)

```
streamlit run dashboard/app.py
```

This opens your browser automatically. Click **"Run AuraSCM Merge Analysis"** and you'll see all 5 modules execute live with a final MERGE ALLOWED/BLOCKED verdict.

8. Every time you reopen a terminal to work on this later, you must reactivate the venv first:

```
cd path\to\AuraSCM
venv\Scripts\activate
```

Then commands work normally with plain python.